EMNLP 2026

# Less is MoE: Trimming Experts in Domain-Specialist Language Models

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Paper Code

## KEY FINDINGS

**01 Fisher Metrics locates domain-critical area in the model.**

Removing 12 out of 2.03M dimensions disrupts math generation.

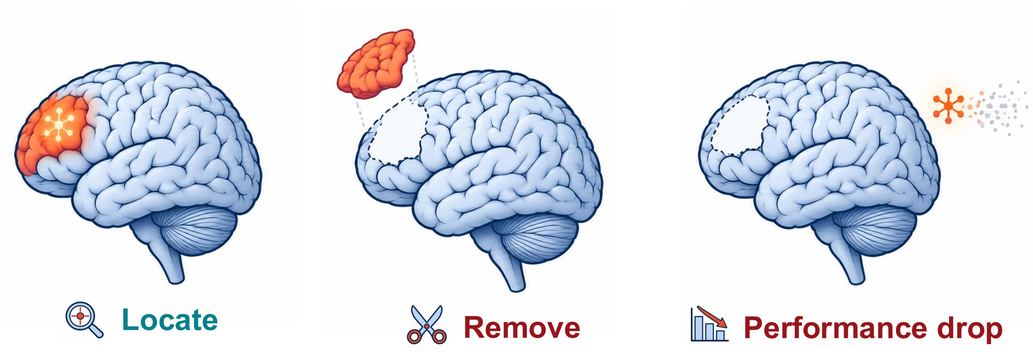
**02 Fisher-MoE beats other pruning baselines.**

Best performance among all 25%, 50% and 75% pruning ratios.

**03 Our project supports auto-integration with vLLM on Fisher-MoE.**

Scan the QR code in the upper-right corner to explore our GitHub repository.

Highlighted region: domain-specific related area in the MoE models




## I. Pruning at expert level with Fisher Importance Metrics

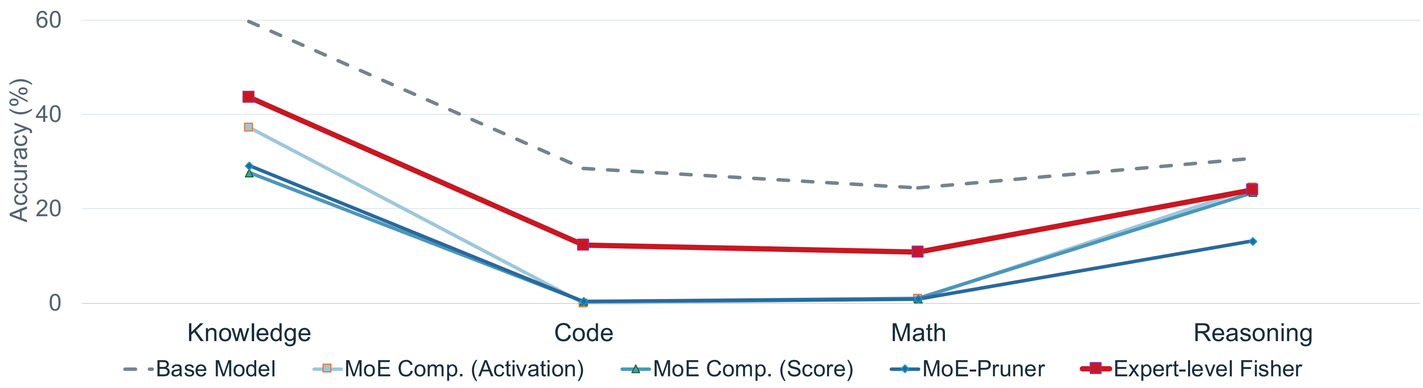
### Fisher Importance Metrics

$$s_{i,j}^{\text{Fisher}} = \underbrace{\frac{1}{d}\sum_{k}}_{\text{Average}} \Big[ \underbrace{(G_i^g)_{j,k}^2}_{\text{Gate gradients}} + \underbrace{(G_i^u)_{j,k}^2}_{\text{Up gradients}} + \underbrace{(G_i^d)_{k,j}^2}_{\text{Down gradients}} \Big]$$

Importance

$$\underbrace{G_i^m}_{\text{Gradient}} := \underbrace{\nabla_{W_i^m}}_{\text{Weight derivative}} \overbrace{\mathcal{L}}^{\text{Calibration loss}}, \underbrace{m \in \{g,u,d\}}_{\text{Projection type}}$$

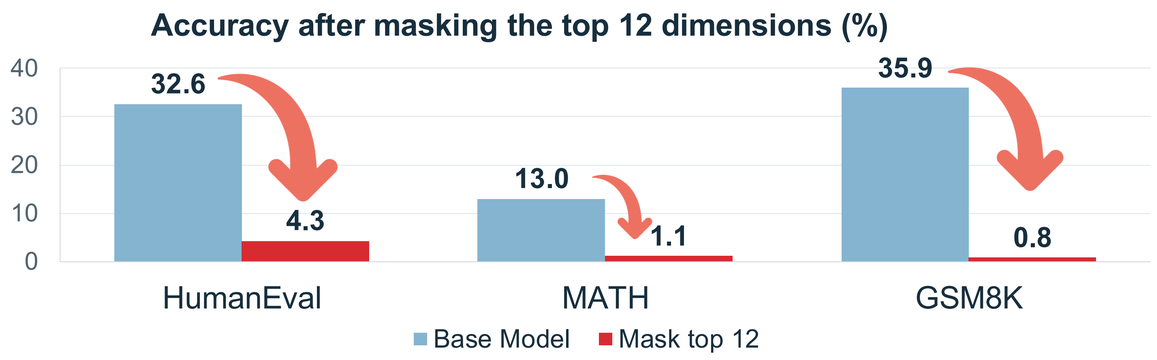
**i** Expert
**j** FFN dimension
**k** Hidden coordinate
**d** Hidden size

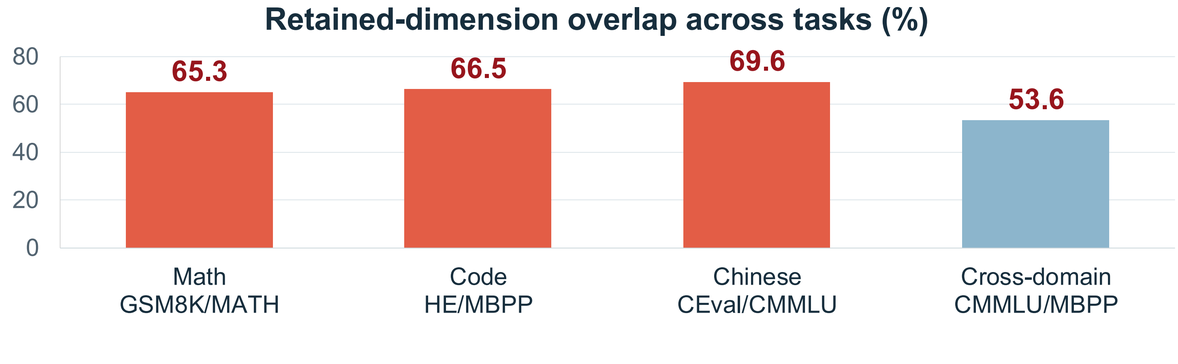



**Expert-level Fisher better preserves knowledge, code, and math accuracy!**

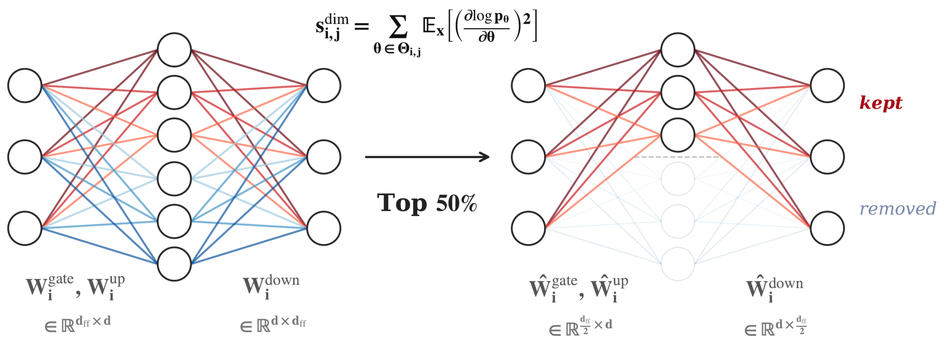
## II. Fisher metrics locates domain-critical parameters

**12 / 2.03M = 0.0006%**



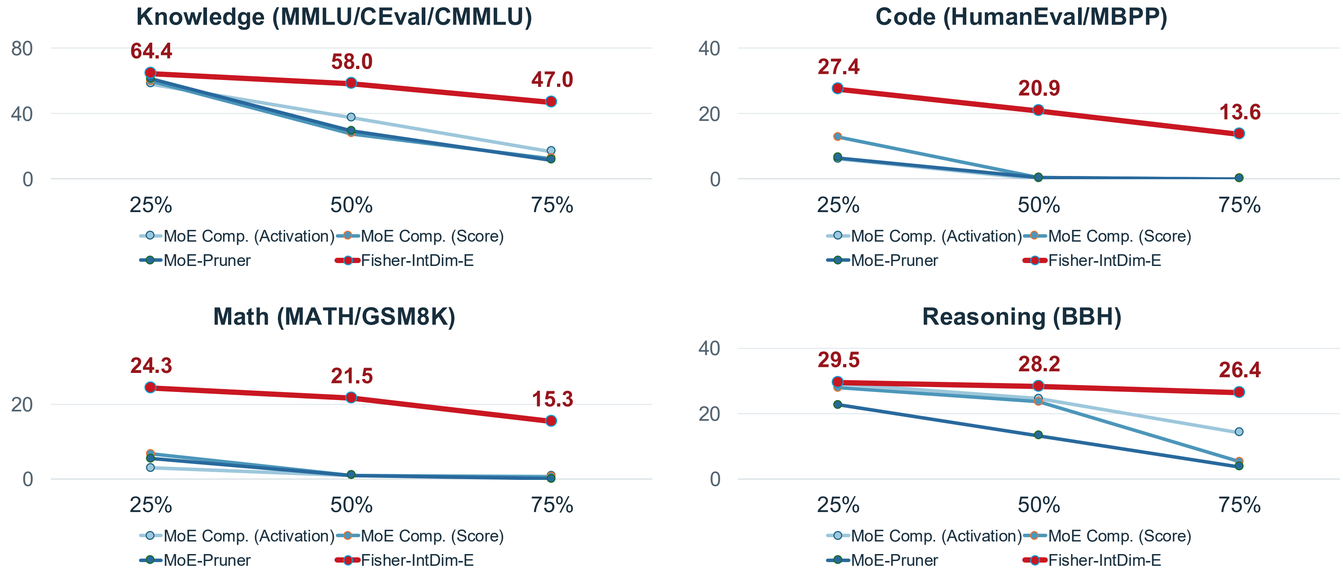

## III. Finer compression granularity with kernel auto-adaptation




**Keep every expert. Remove low-importance FFN dimensions.**

### Implementation

- Automatically detect the **MoE layout**.
- Automatically select and optimize the **vLLM adapter and GPU kernels** (**v0.29.0**).
- **Models:** OLMoE, Qwen3-30B-A3B, Qwen3.5-35B/122B, GPT-OSS-120B, and more.




**Highest accuracy in all domains at 25%, 50%, and 75% pruning.**

## IV. Fisher-MoE preserves better performance comparing other MoE pruning baselines

50% compression for all methods. Experiments also cover OLMoE and Qwen3.5.

| Method | Qwen1.5-MoE-A2.7B | | | | | | | | | | Qwen3-30B-A3B | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | MMLU | CEval | CMMLU | HE | MBPP | MATH | GSM8K | MA | BBH | Avg. | AIME25 | AIME26 | Olympiad | MATH-500 | GSM8K | GPQA-D | Avg. |
| Base Model | 59.3 | 59.3 | 60.7 | 32.6 | 24.6 | 13.0 | 35.9 | 76.7 | 30.7 | 43.6 | 33.3 | 50.0 | 49.6 | 64.2 | 61.9 | 48.5 | 51.3 |
| MoE Comp. (Activation) | 26.3 | 42.5 | 43.1 | 0.0 | 0.0 | 0.0 | 1.9 | 3.3 | 24.5 | 15.7 | 0.0 | 0.0 | 1.2 | 1.6 | 2.8 | 2.5 | 1.4 |
| MoE Comp. (Score) | 37.2 | 13.5 | 32.3 | 0.0 | 0.8 | 0.0 | 1.9 | 2.0 | 23.5 | 12.4 | 0.0 | 0.0 | 0.9 | 1.8 | 2.3 | 10.6 | 2.6 |
| MoE-Pruner | 18.6 | 33.3 | 35.8 | 0.8 | 0.0 | 0.1 | 1.7 | 2.7 | 13.2 | 11.8 | 0.0 | 0.0 | 0.3 | 2.4 | 0.8 | 5.1 | 1.4 |
| **Expert-level Fisher** | 37.9 | 48.6 | 44.2 | 12.9 | 11.7 | 2.4 | 18.0 | 37.7 | 24.1 | 26.4 | 36.7 | 53.3 | 54.9 | **76.2** | **93.4** | 49.0 | 60.6 |
| **Fisher-IntDim-G** | **49.0** | **61.3** | **64.9** | **34.2** | **24.1** | **9.8** | **27.2** | **92.2** | **28.2** | **43.4** | **60.0** | **56.7** | **58.6** | 74.6 | 81.4 | **50.0** | **63.5** |

Accuracy (%). HE: HumanEval. MA: MultiArith. GPQA-D: GPQA-Diamond. E/L/G denote allocation per expert / per layer / globally.

### Analysis from other experiments

- **Performance Recovery:** With half the FFN parameters, domain fine-tuning the pruned model using Fisher-MoE recovers accuracy close to the uncompressed Base + SFT models.
- **Out of domain transfer:** By using the same calibration set, Fisher-MoE retains better performance using out-of-domain calibration dataset.
- **If you want to have a smaller model:** Structural pruning composes with 4-bit quantization and further reduces the model footprint.

### Deployment on one H100 with Qwen1.5-MoE-a2.7B-14B

MODEL VRAM

**↓43.4%**

**26.67 → 15.09 GiB**

THROUGHPUT

**↑21%**

**7,336 → 8,909 tok/s**